

ECO114 Unit 7 Part A Final Boss 80%+

The firm and its customers / profit maximisation: product differentiation, demand, costs, marginal revenue, $MR = MC$, economies of scale, monopoly and price-setting firms.

Part A only Interactive progress Printable MCQ + TF training Graph bank



Dashboard + source audit

Section progress0 / 3 (0%)

- ☐ I understand this is Unit 7 Part A only, not the full Unit 7–8 combined deck.
- ☐ I can state the A2 mark structure and why MCQ/TF precision matters.
- ☐ I know which later topics are excluded from this Part A pack.

ECO114 Unit 7 Part A — The Firm and Its Customers / Profit Maximisation

This pack builds only the **Part A → Unit 7 core**: product differentiation, demand, profit, cost functions, marginal revenue, $MR = MC$, economies/diseconomies of scale, monopolies, and price-setting firms. Unit 8 perfect competition, Part C welfare/fairness, Part D elasticity, and Part E taxes/price controls remain separate packs.

A2: 100 marks 3 hours TF 16 marks with negative marking

MCQ 44 marks with negative marking Longer questions 40 marks

Sources used

- 2026 Econ 114 Unit 7 and 8 Part A slide PDF.
- Existing Economics Unit 7 MegaNote HTML as the prior base.
- CORE Econ Micro Unit 7 sections: 7.1, 7.2, 7.3, 7.4 and 7.6 as the textbook backbone.
- A2 structure slide for exam format.

Important scope boundary

This is **not** the full Unit 7 final pack yet. It is the Unit 7 Part A core pack. Later we still need:

- Part C bridge: surplus, efficiency, fairness, deadweight loss.
- Part D: price elasticity of demand.
- Possible Part 6/competition policy if your lecturer adds it.

Minimum 80% standard

You should not move on until you can solve a new price-setting firm graph by: reading demand, deriving/using MR, finding Q where MR = MC, reading P from demand, calculating profit, and explaining why $P > MC$ for a price-setting firm.

1. Big picture: firms, customers and price setting

Section progress

0 / 3 (0%)

- ☐ I can explain how Unit 7 moves from firm-worker decisions to firm-customer decisions.
- ☐ I can explain why price-setting firms face a trade-off between price and quantity.
- ☐ I can distinguish price-setting firms from price-taking firms.

Unit 6 focused on the firm's relationship with workers: wages, effort, employment rents and labour discipline.
Unit 7 Part A switches to the firm's relationship with customers: price, quantity, product features, cost and profit.

Firm decision	Unit 6 side	Unit 7 Part A side
Who does the firm interact with?	Workers/employees	Customers/buyers
Main choice variable	Wage and employment	Price and quantity
Main constraint	No-shirking/reservation wage logic	Demand curve and cost function
Main profit logic	Hire/pay to maximise profit	Choose Q where $MR = MC$, then set P from demand

Core chain

Differentiated product → market power → downward-sloping demand → firm is a price setter → MR lies below demand/price → profit-maximising Q where $MR = MC$ → price read from demand curve.

Price-setting versus price-taking

Feature	Price-setting firm	Price-taking firm
Product	Differentiated or monopoly product	Homogeneous/identical product
Market power	Some or strong	None individually
Demand faced by individual firm	Downward-sloping	Perfectly elastic at market price
Can set price above rivals?	Yes, but loses some customers	No, loses all customers
Profit rule	MR = MC; P from demand	Usually P = MC in competitive equilibrium

▼ Quick check: Why can Apple charge more than a generic phone producer?

Because Apple sells a differentiated product with brand, design, ecosystem and perceived quality differences. These reduce substitutability and give Apple some market power.

2. Markets + product differentiation

Section progress0 / 3 (0%)

☐ I can define a market in economics.

☐ I can define homogeneous and differentiated goods.

☐ I can explain why differentiation creates market power.

Markets

A market brings buyers and sellers together so that they can transact. In economics, a market is usually defined around one good or service, not necessarily a physical place. The crude oil market, market for phones, and market for breakfast cereal can all be markets even when trades happen through many channels.

Product differentiation

Type	Meaning	Firm behaviour	Exam example
Homogeneous goods	Products are identical in all buyer-relevant ways.	Firms are price takers because buyers will switch fully if one seller charges more.	Very similar agricultural commodities; identical inputs.
Differentiated goods	Products differ by quality, style, brand, location, service, features, convenience or perceived status.	Firms can set their own price because a higher price loses some, but not all, customers.	Apple, breakfast cereals, shoes, restaurants, Shoprite vs spaza/convenience options.

Common trap

Do not say a differentiated firm has unlimited power. It has *some* market power, but it is still constrained by demand and by competitors/substitutes.

Why this matters for Part A

Product differentiation is the reason the individual firm's demand curve slopes downward. That downward-sloping demand curve is the entire foundation of the MR and profit-maximisation model.

▼ **Application: A coffee shop raises its price by R5 but does not lose all customers. What does that tell you?**

It suggests the coffee shop is not a pure price taker. Its location, brand, taste, convenience or atmosphere gives it some differentiation and therefore some market power.

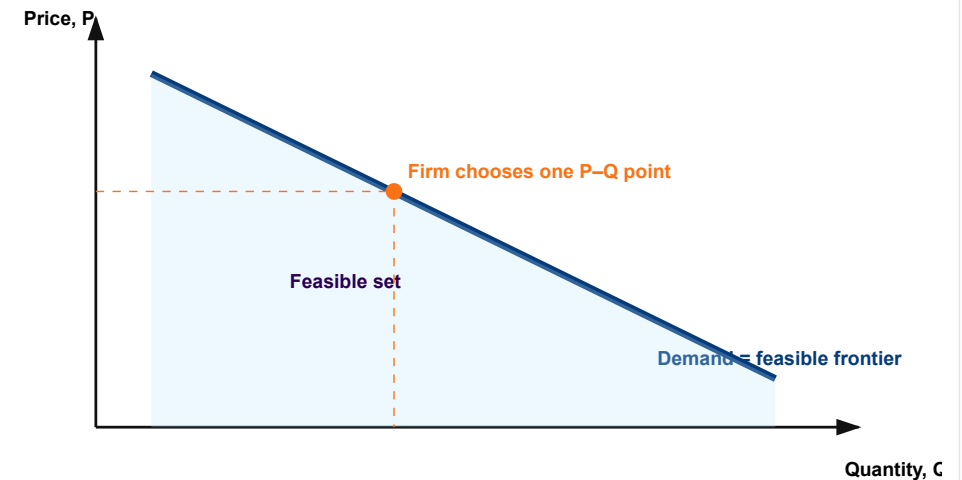
3. Demand, WTP and the law of demand

Section progress

0 / 3 (0%)

- ☐ I can explain demand as feasible P–Q combinations for the firm.
- ☐ I can explain the law of demand using WTP, substitutes and affordability.
- ☐ I can build total demand from individual consumers.

Demand means that buyers **want** the product, **can afford** it, and **plan to buy** it. For a price-setting firm, the demand curve shows all feasible price and quantity combinations.



A price-setting firm cannot choose any price and quantity it likes. Demand shows feasible combinations: higher P means lower Q demanded.

Law of demand

The law of demand says that there is an inverse relationship between price and quantity demanded: as price rises, quantity demanded falls; as price falls, quantity demanded rises.

Why demand slopes down	Explanation
Diminishing marginal utility / WTP	As a person consumes more units, the extra satisfaction from another unit usually falls, so WTP for additional units falls.
Substitutes	If price rises, some consumers switch to competitor products or alternatives.
Affordability / income effect	For expensive products, a higher price can make fewer consumers able or willing to buy.
Indivisible goods	For large purchases like cars, a price rise may cause consumers not to buy at all.

Individual and total demand

Each consumer's WTP schedule can be treated as an individual demand curve. Total demand is found by adding quantities demanded across consumers at each price.

At a given price: Total quantity demanded = $Q_A + Q_B + Q_C + \dots$

Slide-style exercise method

If a table gives consumers A, B and C quantity demanded at each price, do not average their quantities. Add them horizontally at the same price.

Movement along demand vs shift in demand

Event	Graph effect	Example
Own price changes	Movement along demand	The firm cuts price from R50 to R40, so Q demanded rises along the same curve.
Preference/income/competitor/quality changes	Demand curve shifts	Advertising improves perceived quality, so at every price consumers demand more.

▼ Quick check: Does a firm's own price increase shift its demand curve left?

No. The firm's own price change causes movement along the demand curve. Demand shifts when something other than own price changes, such as preferences, income, competitor prices, quality, new markets or advertising.

4. Profit framework: revenue, cost and profit

Section progress0 / 3 (0%)

- ☐ I can write and use profit = TR – TC.
- ☐ I can explain why the firm wants high P and high Q but cannot choose both freely.
- ☐ I understand the note that CORE isoprofit curves are not assessed in ECO114 Part A.

The firm's objective in this model is to maximise profit. Profit depends on revenue from customers and production costs.

Profit: $\pi = TR - TC$
Total revenue: $TR = P \times Q$
With constant unit cost c: $TC = c \times Q$
Therefore: $\pi = (P - c) \times Q$

It might look as if the firm should simply choose a high price and high quantity. The problem is that demand prevents this: a high price reduces quantity demanded. The firm must choose the best feasible combination on the demand curve.

Isoprofit note

The CORE text uses isoprofit curves as a firm version of indifference curves: each curve joins P–Q combinations with equal profit. Your Part A slides explicitly say these are **not included in Economics 114 assessment**. You should understand the intuition, but prioritise profit tables and MR = MC.

Exam trap

Do not maximise profit by choosing the highest price. A very high price may produce very low quantity and therefore low total profit.

▼ If P = R150, Q = 500 and unit cost c = R60, calculate profit.

$\pi = (P - c) \times Q = (150 - 60) \times 500 = \text{R}45\,000.$

5. Cost functions: TC, AC, MC and fixed costs

Section progress

0 / 3 (0%)

☐ I can define total cost, average cost and marginal cost.

☐ I can calculate AC and MC from a table.

☐ I can identify fixed cost from TC when Q = 0.

Costs are not always constant per unit. As output changes, firms may use fixed inputs more or less efficiently, encounter capacity problems, or benefit from specialisation.

Total cost: $TC = TC(Q)$
Average cost: $AC = TC(Q) / Q$
Marginal cost: $MC = \Delta TC / \Delta Q$ or dTC/dQ
Fixed cost = TC when $Q = 0$

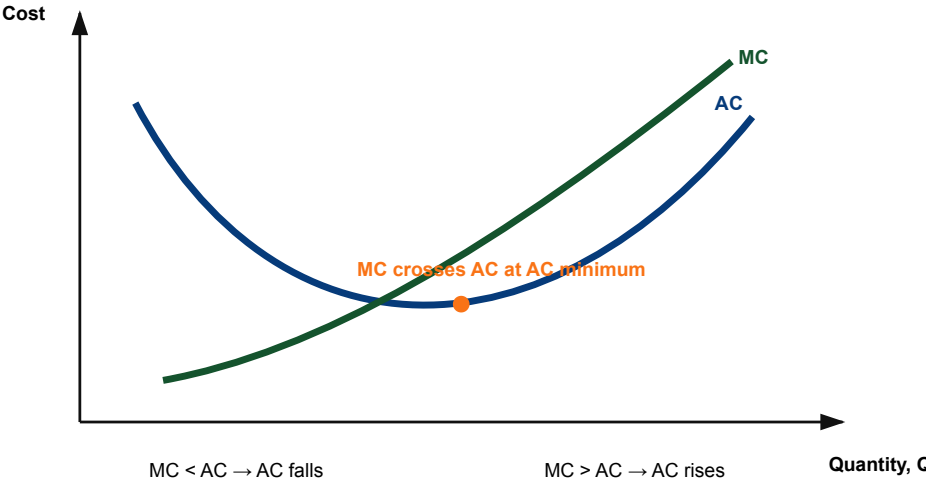
Constant unit-cost example

If unit cost is R60 per chair, then $TC = 60Q$, $AC = R60$ for every positive Q , and $MC = R60$. At $Q = 0$, AC is undefined because you cannot divide by zero.

Beautiful Cars style table

Q	TC	AC = TC/Q	MC idea
0	48 000	Undefined	Undefined
20	80 000	4 000	Approx. 2 200
40	136 000	3 400	Approx. 3 400
60	216 000	3 600	Approx. 4 600

The fixed cost is **48 000**, because that is the cost at $Q = 0$.



Average cost often falls at low output and rises at high output. Marginal cost pulls average cost down when below AC and up when above AC.

Graph language

When MC is below AC, the next unit costs less than the current average, so AC falls. When MC is above AC, the next unit costs more than the current average, so AC rises.

6. Cost maths worked examples

Section progress0 / 3 (0%)

- ☐ I can derive AC from TC.
- ☐ I can derive MC from TC using differentiation.
- ☐ I can find the Q where AC is minimised in a quadratic-cost example.

Example 1: $TC(Q) = 1100 + 140Q + 0.2Q^2$

$AC = TC/Q = 1100/Q + 140 + 0.2Q$
 $MC = dTC/dQ = 140 + 0.4Q$

Example 2: Slide-style exercise

Given: $TC(Q) = 62\,500 + 500Q + Q^2$.

$MC = dTC/dQ = 500 + 2Q$
 $MC \text{ at } Q = 1050: 500 + 2(1050) = 2\,600$

To find where AC is minimised:

$AC = 62\,500/Q + 500 + Q$
 $dAC/dQ = -62\,500/Q^2 + 1$
Set $dAC/dQ = 0 \rightarrow Q^2 = 62\,500 \rightarrow Q = 250$

Calculation trap

For AC, divide the whole TC function by Q. Do not divide only the variable-cost terms. Fixed cost becomes fixed cost divided by Q.

▼ Why does fixed cost make AC high at low Q?

Because the fixed cost is spread over very few units. As Q rises, fixed cost per unit falls, pushing AC down at first.

7. Marginal revenue: the gain/loss logic

Section progress0 / 3 (0%)

- ☐ I can define MR as $\Delta TR/\Delta Q$.
- ☐ I can explain why MR is below price for a downward-sloping demand curve.
- ☐ I can do the Q = 19 to Q = 20 revenue calculation.

Marginal revenue is the change in total revenue when quantity rises by one unit, or the slope of the total revenue function.

$MR = \Delta TR / \Delta Q$
or, using calculus: $MR = dTR/dQ$

Why MR is below price

When a price-setting firm sells one extra unit, two things happen:

1. **Output effect:** it gains revenue from selling the extra unit.
2. **Price effect:** it must lower the price on all units, including the units it was already selling.

Because the price effect subtracts revenue, MR is less than the price.

Beautiful Cars slide example

Change	Value
At Q = 19, P = \$6 480	$TR = 19 \times 6\,480 = \$123\,120$
At Q = 20, P = \$6 400	$TR = 20 \times 6\,400 = \$128\,000$
MR	$128\,000 - 123\,120 = \$4\,880$
Gain/loss method	Gain = \$6 400; loss = $19 \times \$80 = \$1\,520$; $MR = \$6\,400 - \$1\,520 = \$4\,880$

If demand is linear: $P = a - bQ$
 $TR = P \times Q = aQ - bQ^2$
 $MR = a - 2bQ$
MR has the same vertical intercept as demand, but twice the slope.

Exam trap

Never read the firm's price from the MR curve. MR helps choose Q. Demand gives the price customers will pay for that Q.

8. Profit maximisation: $MR = MC$, then price from demand

Section progress

0 / 3 (0%)

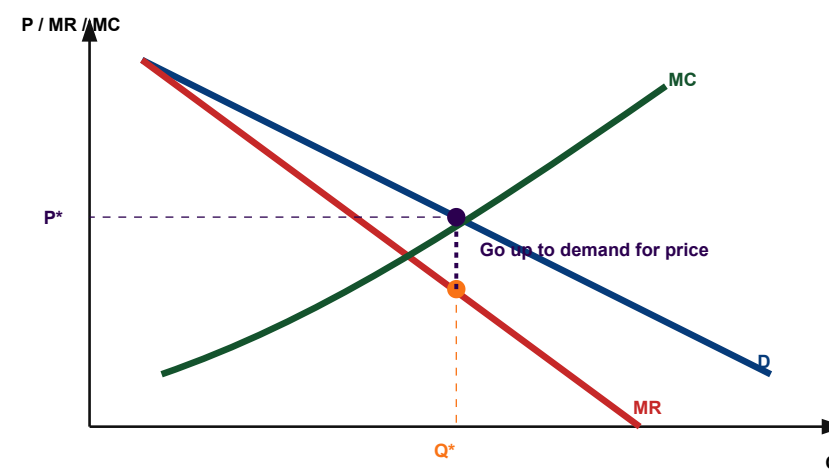
- ☐ I can explain why $MR > MC$ means produce more.
- ☐ I can explain why $MR < MC$ means produce less.
- ☐ I can use $MR = MC$ to find Q^* and demand to find P^* .

The firm maximises profit where the extra revenue from one more unit equals the extra cost of one more unit.

$$\text{Profit: } \pi = TR - TC$$

$$d\pi/dQ = dTR/dQ - dTC/dQ = MR - MC$$

$$\text{Profit maximised when } MR - MC = 0 \rightarrow MR = MC$$



Profit-maximising quantity is where $MR = MC$. The price is not read from MR; it is read from the demand curve at Q^* .

Condition	Decision	Reason
MR > MC	Increase output	The extra unit adds more to revenue than to cost.
MR < MC	Decrease output	The extra unit adds more to cost than to revenue.
MR = MC	Profit-maximising Q	No nearby output change can increase profit.

Step-by-step exam method

1. Write down demand, TR, MR and MC if given.
2. Set MR = MC.
3. Solve for Q*.
4. Substitute Q* into the demand equation to find P*.
5. Calculate profit using $\pi = TR - TC$ or $\pi = Q(P - AC)$.
6. Interpret the result in words.

▼ Demand is $P = 8000 - 80Q$ and $MC = 2880$. Find Q^* and P^* .

MR = $8000 - 160Q$. Set MR = MC: $8000 - 160Q = 2880 \rightarrow 160Q = 5120 \rightarrow Q^* = 32$. Price: $P = 8000 - 80(32) = 5440$.

9. Economies and diseconomies of scale

Section progress

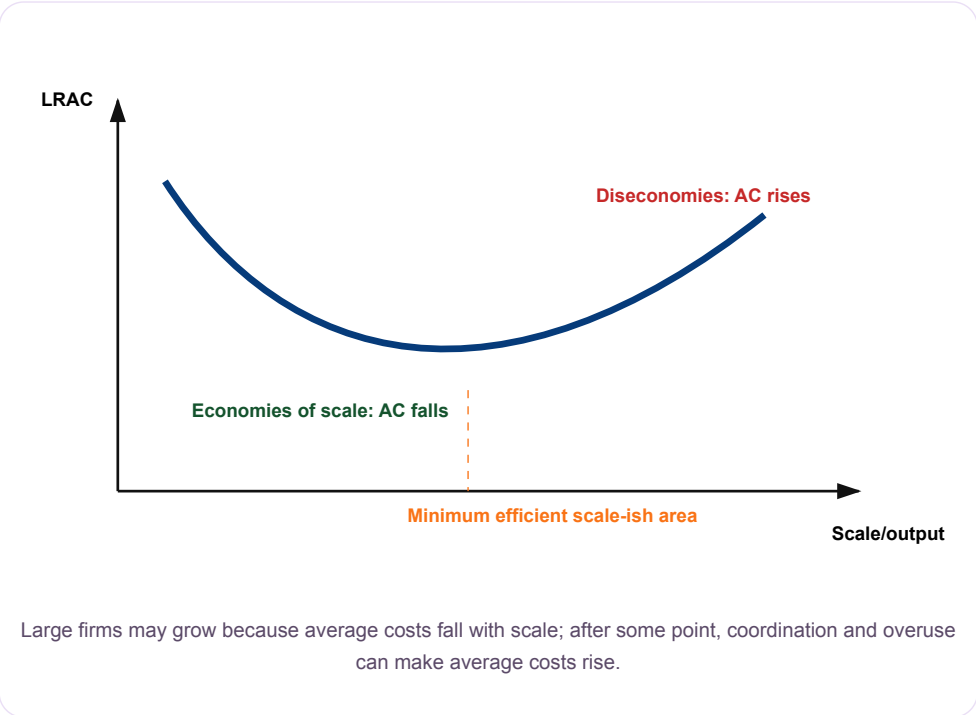
0 / 3 (0%)

- ☐ I can define economies of scale.
- ☐ I can define diseconomies of scale.
- ☐ I can explain technological, cost and network economies.

Economies of scale help explain why some firms become very large. They occur when increasing all inputs by a proportion increases output by a larger proportion, causing long-run average cost to fall.

Source of scale advantage	Meaning	Example logic
Technological advantages	Large-scale production uses fewer inputs per unit.	Large power stations or production lines can be more efficient than tiny units.
Cost advantages	Large firms buy inputs at lower prices.	Shoprite can bargain for lower supplier prices than a small spaza shop.
Network economies	The product becomes more valuable because many others use it.	Platforms, apps and social networks become more useful as user numbers rise.

Diseconomies of scale occur when increasing all inputs by a proportion increases output by a smaller proportion. This can happen due to over-complex management, congestion, supervision problems, coordination failures or breakdowns.



Likely longer-question angle
Explain why large firms exist: connect economies of scale to falling average costs, then give at least two sources such as technology, input bargaining power and network effects.

10. Monopolies and price-setting firms

Section progress0 / 3 (0%)

☐ I can distinguish monopolistic competitors from monopolies.

☐ I can explain barriers to entry.

☐ I can calculate monopoly profit using $Q(P - AC)$.

Differentiated firms vs monopolies

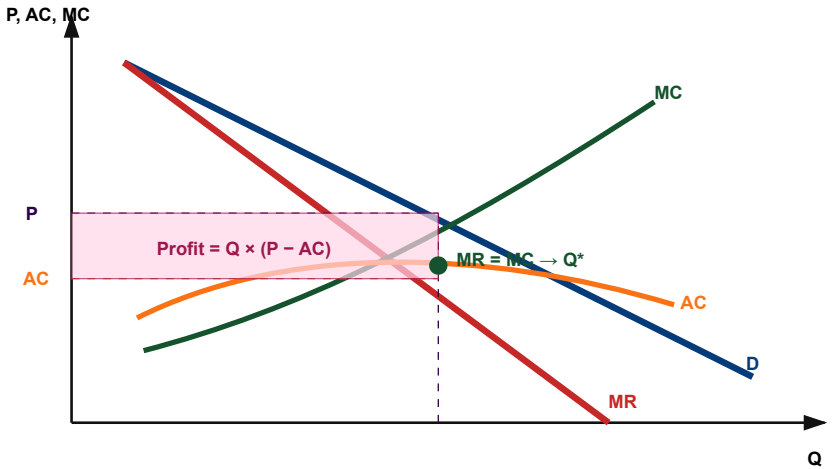
Feature	Differentiated firm / monopolistic competitor	Monopoly
Number of sellers	Many firms selling varieties of similar products	One firm controls market supply
Substitutes	Some substitutes exist	No close substitutes
Demand	Downward-sloping firm demand	Downward-sloping market demand
Market power	Some	Strong
Graphical method	$MR = MC$, price from demand	Same method, but demand is market demand

Barriers to entry

- **Exclusive ownership of a resource:** control over key inputs.
- **Legal barriers:** patents, copyright, licences or regulation.
- **Cost barriers / economies of scale:** natural monopoly where one large producer has lower average cost than multiple small producers.

Markup and profit

Price-setting firm at optimum: $P > MR = MC \rightarrow P > MC$
Profit margin / markup in rand terms: $P - MC$
Profit margin ratio: $(P - MC) / P$
Monopoly profit: $\pi = TR - TC = P \times Q - AC \times Q = Q(P - AC)$



Monopoly/price-setting profit can be shown as a rectangle: quantity times the gap between price and average cost.

Monopoly trap

A monopoly does not choose the highest price possible. It chooses the price and quantity combination that maximises profit, constrained by market demand.

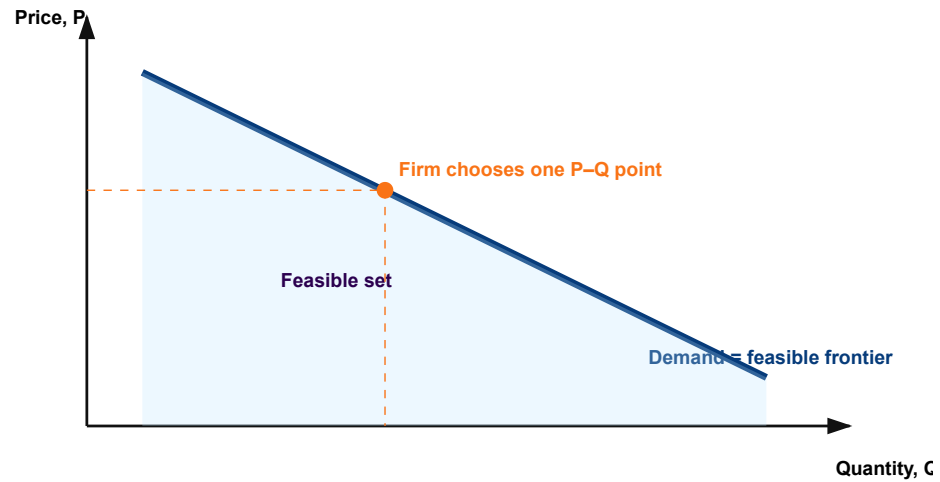
11. Graph and diagram bank

Section progress

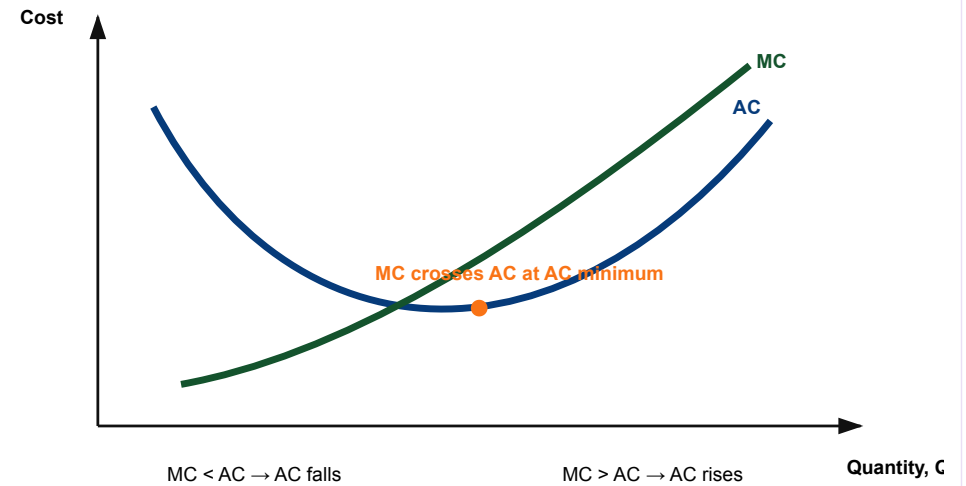
0 / 3 (0%)

- ☐ I can draw and label every Part A diagram from memory.
- ☐ I know what the axes mean on each graph.
- ☐ I can write the exam explanation below each graph.

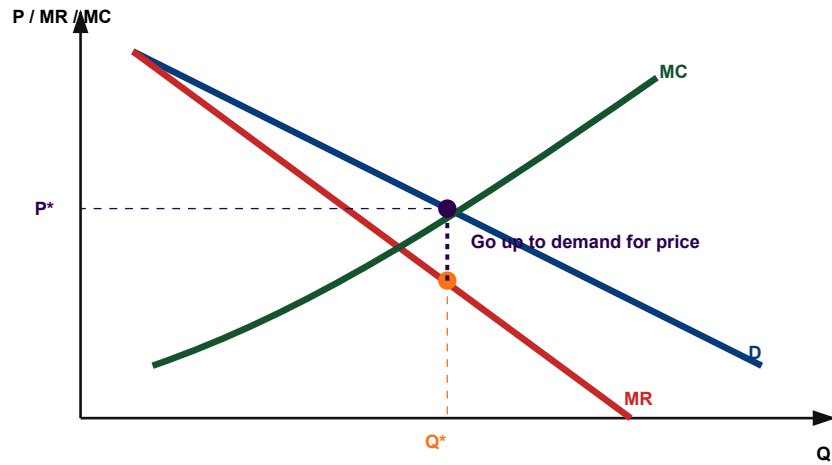
Graph	Axes	How to draw	Exam sentence
Demand curve	Price on y-axis; quantity on x-axis	Downward-sloping line/curve	Demand shows feasible P–Q combinations and WTP for marginal units.
TC curve	Total cost on y-axis; Q on x-axis	Starts at fixed cost if fixed costs exist; rises as Q rises	TC shows the total cost of producing each output level.
AC curve	Average cost on y-axis; Q on x-axis	Often U-shaped	AC falls when fixed costs are spread and rises when capacity/coordination problems appear.
MC curve	Marginal cost on y-axis; Q on x-axis	Often upward-sloping or U-shaped	MC is the extra cost of one more unit and crosses AC at the AC minimum.
Demand + MR + MC	Price/MR/MC on y-axis; Q on x-axis	D slopes down, MR below D, MC upward	Q* is where MR = MC; P* is read from demand at Q*.
LRAC and scale	Average cost on y-axis; scale/output on x-axis	Falls under economies, rises under diseconomies	Scale can explain why some firms become large and why natural monopoly may occur.



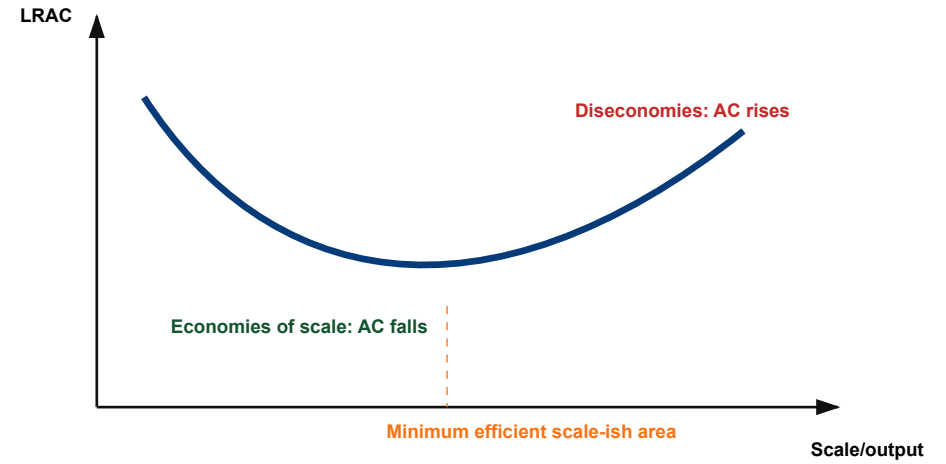
A price-setting firm cannot choose any price and quantity it likes. Demand shows feasible combinations: higher P means lower Q demanded.



Average cost often falls at low output and rises at high output. Marginal cost pulls average cost down when below AC and up when above AC.



Profit-maximising quantity is where $MR = MC$. The price is not read from MR; it is read from the demand curve at Q^* .



Large firms may grow because average costs fall with scale; after some point, coordination and overuse can make average costs rise.

12. Formula and calculation bank

Section progress0 / 3 (0%)

☐ I can use every Part A formula correctly.

☐ I can avoid confusing AC with MC.

☐ I can interpret final answers in words.

Formula	Meaning	Trap
$TR = P \times Q$	Total money received from sales	Price may depend on Q through demand.
$\pi = TR - TC$	Profit	Do not forget fixed costs inside TC.
$AC = TC/Q$	Cost per unit on average	Undefined at $Q = 0$.
$MC = \Delta TC/\Delta Q$ or dTC/dQ	Extra cost from extra output	MC is not TC/Q .
$MR = \Delta TR/\Delta Q$ or dTR/dQ	Extra revenue from extra output	For a price setter, $MR < P$.
If $P = a - bQ$, $MR = a - 2bQ$	Linear-demand shortcut	Only applies directly to linear demand.
Profit max: $MR = MC$	Choose Q^*	Read P from demand, not MR.
$\pi = Q(P - AC)$	Profit rectangle	Use AC, not MC, for total profit rectangle.

Worked MR exercise from slides

Demand: $Q = 100 - P/80$. Rearranged: $P = 8000 - 80Q$.

$$TR = P \times Q = (8000 - 80Q)Q = 8000Q - 80Q^2$$
$$MR = dTR/dQ = 8000 - 160Q$$

From $Q = 10$ to $Q = 11$:

$$P_{10} = 7200; TR_{10} = 72\ 000$$
$$P_{11} = 7120; TR_{11} = 78\ 320$$

$$MR = 78\ 320 - 72\ 000 = 6\ 320$$
$$\text{Gain} = 7\ 120; \text{loss} = 10 \times 80 = 800; MR = 7\ 120 - 800 = 6\ 320$$

13. True/False bank with traps

Section progress

0 / 3 (0%)

☐ I can answer these without reckless guessing.

☐ I can explain every false statement.

☐ I know which words make statements dangerous.

Use this for negative-marking training. Cover the answer, decide, then reveal.

▼ 1. A firm selling a differentiated product usually faces a downward-sloping demand curve.

True. Differentiation means some consumers remain even if price rises, but quantity demanded falls.

▼ 2. A price-setting firm can choose any price and quantity combination it wants.

False. Demand constrains feasible P–Q combinations.

▼ 3. A homogeneous good gives each firm more market power than a differentiated good.

False. Homogeneous goods usually reduce individual market power.

▼ 4. If the firm raises its own price, this causes a movement along its demand curve rather than a shift of the curve.

True. Own price changes move along demand.

▼ 5. Advertising that changes perceived quality can shift a firm's demand curve.

True. This changes demand at every price.

▼ 6. Average cost is total cost divided by quantity.

True. $AC = TC/Q$.

▼ 7. Marginal cost is total cost divided by quantity.

False. That is average cost; MC is the extra cost of extra output.

▼ 8. If $Q = 0$, average cost is undefined.

True. Division by zero is undefined.

▼ 9. Fixed cost can be identified as total cost when output is zero.

True. Variable costs are zero at $Q=0$, so remaining cost is fixed cost.

▼ 10. When MC is below AC, AC is falling.

True. A below-average extra unit pulls the average down.

▼ 11. When MC is above AC, AC is falling.

False. A higher-than-average extra unit pulls the average up.

▼ 12. For a price-setting firm with downward-sloping demand, MR is usually below price.

True. Selling more requires lowering price on previous units too.

▼ 13. If MR exceeds MC, producing one more unit increases profit.

True. Extra revenue exceeds extra cost.

▼ 14. The profit-maximising price-setting firm chooses the quantity where price equals MC.

False. It chooses Q where $MR = MC$; P is read from demand.

▼ 15. For linear demand $P = a - bQ$, marginal revenue is $MR = a - 2bQ$.

True. MR has the same intercept and twice the slope.

▼ 16. A monopoly's demand curve is the market demand curve.

True. The monopolist is the only seller in the market.

▼ 17. A monopoly always charges the maximum price consumers could possibly pay.

False. It chooses the price–quantity combination that maximises profit.

▼ 18. Barriers to entry help protect monopoly power.

True. They prevent or limit entry by competitors.

▼ 19. Economies of scale mean that average cost rises as output expands.

False. Economies of scale mean average cost falls as scale expands.

▼ 20. Diseconomies of scale can arise from management and coordination problems.

True. Very large scale can create supervision, congestion and complexity issues.

▼ 21. Network economies occur when a product is more valuable because many people use it.

True. This is common in platforms and digital services.

▼ 22. Profit can be calculated as $Q(P - AC)$.

True. Since $TC = AC \times Q$, profit = $P \times Q - AC \times Q$.

▼ 23. Profit can be calculated as $Q(P - MC)$ in the standard profit rectangle.

False. Use AC for total profit rectangle, not MC.

▼ 24. Product differentiation means consumers never switch to competitors.

False. Some may switch; not all.

▼ 25. The demand curve can be interpreted as willingness to pay for the marginal unit.

True. Demand can show WTP for each quantity.

▼ 26. Total demand is found by horizontally adding individual demand quantities at each price.

True. Add quantities, not prices.

▼ 27. The CORE isoprofit-curve method is the central assessed ECO114 method in these slides.

False. Slides note isoprofit curves are not included in Economics 114 assessment.

▼ 28. A natural monopoly may arise from strong economies of scale.

True. One large firm may produce at lower average cost than many small firms.

▼ 29. A price-setting firm with market power is still constrained by demand.

True. Market power is not unlimited power.

▼ 30. If demand shifts right, consumers demand more at every price.

True. A rightward shift means more quantity at the same price.

▼ 31. If price falls and quantity demanded rises, this is always a shift of demand.

False. It is usually a movement along demand if own price changed.

▼ 32. MR can be negative for a price-setting firm.

True. If price cuts on existing units exceed revenue from the extra unit, MR is negative.

▼ 33. At the profit maximum, total revenue must be maximised.

False. Profit maximisation depends on revenue and cost, not revenue alone.

▼ 34. A firm with lower average cost can sometimes profit at a lower price than rivals.

True. Cost advantages can support low-price strategies.

▼ 35. Monopolistic competitors and monopolies are both price setters.

True. Both face downward-sloping demand, though monopoly power is stronger.

14. MCQ bank with explanations

Section progress

0 / 3 (0%)

- ☐ I can eliminate wrong options.
- ☐ I can explain the correct answer.
- ☐ I can identify recurring MCQ traps.

MCQ is the biggest A2 mark bank. Do not just memorise letters; understand why the distractors fail.

▼ 1. A firm sells a branded product and loses some, but not all, customers when it raises price. This firm is best described as:

- A. a pure price taker
- B. a firm with some market power
- C. a firm with no demand curve
- D. a firm that can sell unlimited quantity at any price

Correct answer: a firm with some market power.

Differentiation gives some market power, but demand still constrains the firm.

▼ 2. Which event shifts a firm's demand curve rather than moving along it?

- A. The firm lowers its own price
- B. The firm raises its own price
- C. Consumer preferences improve because of advertising
- D. The firm moves from $Q=20$ to $Q=30$ on the same curve

Correct answer: Consumer preferences improve because of advertising.

Non-price demand determinants shift the curve.

▼ 3. If $TC = 500 + 20Q$, the fixed cost is:

- A. 20
- B. 500
- C. 520
- D. undefined

Correct answer: 500.

Fixed cost equals TC when $Q = 0$.

▼ 4. If $TC = 1000 + 100Q + Q^2$, MC is:

- A. $1000/Q + 100 + Q$
- B. $100 + 2Q$
- C. $1000 + 100Q$
- D. $100 + Q$

Correct answer: $100 + 2Q$.

Differentiate TC with respect to Q.

▼ 5. If $P = 100 - 2Q$, then MR is:

- A. $100 - 2Q$
- B. $100 - 4Q$
- C. $200 - 2Q$
- D. $50 - Q$

Correct answer: $100 - 4Q$.

For linear demand, MR has twice the slope.

▼ 6. A price-setting firm maximises profit where:

- A. $P = MC$
- B. $MR = MC$
- C. $AC = MC$ only
- D. $TR = TC$

Correct answer: $MR = MC$.

Q is chosen where marginal revenue equals marginal cost.

▼ 7. After finding Q^* from $MR = MC$, the price is found from:

- A. the MC curve
- B. the AC curve
- C. the demand curve
- D. the fixed cost curve

Correct answer: the demand curve.

Demand tells what consumers will pay for that quantity.

▼ 8. If $MR > MC$ at current output, the firm should:

- A. decrease output
- B. increase output
- C. shut down automatically
- D. raise fixed cost

Correct answer: increase output.

The next unit adds more revenue than cost.

▼ 9. If MC is below AC , then:

- A. AC is rising
- B. AC is falling
- C. AC is undefined
- D. AC must equal price

Correct answer: AC is falling.

A below-average marginal unit lowers the average.

▼ 10. Which is a source of economies of scale?

- A. Fewer users making a platform less useful
- B. Large firms buying inputs at lower prices
- C. Average cost rising at all output levels
- D. Own price causing movement along demand

Correct answer: Large firms buying inputs at lower prices.

Bulk purchasing is a cost advantage.

▼ 11. A natural monopoly is most likely when:

- A. many firms have identical costs
- B. one large firm has lower average cost due to scale
- C. goods are perfectly homogeneous and entry is free
- D. all firms are price takers

Correct answer: one large firm has lower average cost due to scale.

Strong economies of scale can make one firm efficient.

▼ 12. Which statement about monopoly is correct?

- A. A monopoly faces perfectly elastic demand
- B. A monopoly demand curve is the market demand curve
- C. A monopoly cannot choose price
- D. A monopoly must set $P = MC$

Correct answer: A monopoly demand curve is the market demand curve.

A monopolist is the sole seller, so firm demand equals market demand.

▼ 13. Profit rectangle uses:

- A. $Q(P - MC)$
- B. $Q(P - AC)$
- C. $Q(MR - MC)$
- D. $P - Q$

Correct answer: $Q(P - AC)$.

Profit equals quantity times profit per unit, where profit per unit is $P - AC$.

▼ 14. Which best explains why MR is below price for a downward-sloping demand curve?

- A. Fixed cost rises
- B. The firm must lower the price on previous units to sell an extra unit
- C. MC is always negative
- D. Demand shifts right

Correct answer: The firm must lower the price on previous units to sell an extra unit.

This is the price effect.

▼ 15. At $Q=19$, $P=\$6480$; at $Q=20$, $P=\$6400$. What is the loss on previous units when output rises to 20?

- A. \$80
- B. \$1520
- C. \$6400
- D. \$4880

Correct answer: \$1520.

The 19 previous units each lose \$80 of price: $19 \times 80 = 1520$.

▼ 16. If gain on the 20th unit is \$6400 and loss on previous units is \$1520, MR is:

- A. \$7920
- B. \$6400
- C. \$4880
- D. \$1520

Correct answer: \$4880.

MR = gain – loss = 6400 – 1520.

▼ 17. Which is not a barrier to entry?

- A. Patent protection
- B. Control of a key resource
- C. Strong economies of scale
- D. A movement along demand caused by own price

Correct answer: A movement along demand caused by own price.

That is demand mechanics, not an entry barrier.

▼ 18. Product differentiation can be based on:

- A. quality only
- B. brand only
- C. location only
- D. quality, style, brand, location or perceived characteristics

Correct answer: quality, style, brand, location or perceived characteristics.

Differentiation can take many forms.

▼ 19. If all sellers sell truly identical goods and one seller charges more than the others, the likely result is:

- A. it sells unlimited quantity
- B. it loses buyers to competitors
- C. its demand shifts right
- D. MR becomes higher than price

Correct answer: it loses buyers to competitors.

Homogeneous goods make buyers very willing to switch.

▼ 20. Which statement is the safest exam wording?

- A. Market power means a firm can do anything
- B. A price-setting firm has some ability to set price, but is constrained by demand
- C. Monopolies always create fair outcomes
- D. Differentiated products eliminate substitutes

Correct answer: A price-setting firm has some ability to set price, but is constrained by demand.

This wording is precise and avoids overclaiming.

▼ 21. AC is undefined at Q = 0 because:

- A. TC is always zero
- B. MC is negative
- C. division by zero is undefined
- D. fixed cost disappears

Correct answer: division by zero is undefined.

AC = TC/Q.

▼ 22. If TC at Q=0 is R48 000, fixed cost is:

- A. R0
- B. R48 000
- C. R80 000
- D. Cannot be identified

Correct answer: R48 000.

Fixed cost remains even when output is zero.

▼ 23. Which condition can make AC rise at high output?

- A. underuse of fixed inputs
- B. fixed costs spread over more units
- C. overcrowding and management problems
- D. lower supplier prices

Correct answer: overcrowding and management problems.

These are diseconomies of scale.

▼ 24. Demand as WTP means:

- A. the demand curve shows what consumers are willing to pay for a marginal unit
- B. the demand curve equals MC
- C. WTP rises forever as quantity rises
- D. all consumers have identical preferences

Correct answer: the demand curve shows what consumers are willing to pay for a marginal unit.
Demand can be interpreted as marginal WTP.

▼ 25. Why should a firm not maximise revenue instead of profit?

- A. Costs do not matter
- B. Profit depends on both revenue and cost
- C. Revenue and profit are always identical
- D. MR is always negative

Correct answer: Profit depends on both revenue and cost.
High revenue can still be low-profit if costs are high.

▼ 26. Which graph rule is correct?

- A. MR lies above demand for a price-setting firm
- B. MC always equals AC
- C. MR lies below demand when demand slopes down
- D. Demand is upward-sloping by law

Correct answer: MR lies below demand when demand slopes down.
The price effect makes MR less than price.

▼ 27. Which formula is correct for total revenue when $P = 8000 - 80Q$?

- A. $TR = 8000 - 80Q$
- B. $TR = 8000Q - 80Q^2$
- C. $TR = 8000 - 160Q$
- D. $TR = 80Q^2$

Correct answer: $TR = 8000Q - 80Q^2$.
 $TR = P \times Q$.

▼ 28. Which is closest to the reason Shoprite may have lower unit costs than a spaza shop?

- A. It faces no customers
- B. It has cost advantages from scale and supplier bargaining power
- C. It cannot set prices
- D. It has no fixed costs

Correct answer: It has cost advantages from scale and supplier bargaining power.
Large firms can obtain lower input prices.

▼ 29. If demand shifts right, then at the same price:

- A. quantity demanded is lower
- B. quantity demanded is higher
- C. marginal cost disappears
- D. AC must equal MC

Correct answer: quantity demanded is higher.
A rightward demand shift means more demanded at every price.

▼ 30. A firm with monopoly power still cannot:

- A. choose Q using MR and MC
- B. face a downward-sloping demand curve
- C. ignore customers' willingness to pay entirely
- D. earn profit if P exceeds AC

Correct answer: ignore customers' willingness to pay entirely.
Demand/WTP constrains even monopolies.

15. Longer-question templates

Section progress

0 / 3 (0%)

☐ I can write a 6–10 mark answer using the templates.

☐ I can draw diagrams where needed.

☐ I avoid vague business-language answers.

Explain how product differentiation gives a firm market power.

Define differentiation → explain fewer close substitutes → downward-sloping demand → firm can raise price without losing all customers → still constrained by demand.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Explain why a price-setting firm chooses output where $MR = MC$.

Define MR and MC → if $MR > MC$ profit rises by expanding → if $MR < MC$ profit rises by reducing output → optimum at $MR = MC$ → price from demand.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Use a diagram to show how a monopoly chooses price and quantity.

Draw D, MR, MC → find Q^* at $MR = MC$ → go up to D to find P^* → calculate profit rectangle if AC shown → explain $P > MC$.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Explain why marginal revenue lies below demand for a firm with market power.

Increasing Q requires a lower P → gain from extra unit minus loss on previous units → $MR < P$ → use numerical example if possible.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Explain economies of scale and why large firms exist.

Define economies of scale → AC falls with scale → explain technological/cost/network economies → link to Shoprite/Toyota/Amazon examples → mention possible diseconomies.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Distinguish monopolistic competition from monopoly.

Both are price setters and face downward-sloping demand → differentiated firms have competitors/substitutes → monopoly has one seller/no close substitutes → monopoly demand = market demand.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Explain how to calculate and interpret AC and MC from a cost function.

Define TC, AC, MC → divide TC by Q for AC → differentiate or use $\Delta TC / \Delta Q$ for MC → interpret fixed cost at $Q = 0$ → explain MC crossing AC at minimum.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

Discuss whether monopoly is desirable.

Consumers dislike higher prices → owners like profit → objective welfare requires surplus/deadweight loss analysis, which links to Part C → for Part A focus on price, output and profit mechanics.

Starter: In a price-setting market, the firm faces a downward-sloping demand curve, so its price and output decision must be analysed using demand, revenue and cost rather than price-taking assumptions.

16. Active recall drill + 80% checklist

Section progress

0 / 13 (0%)

- ☐ I can answer the 50 recall questions closed-book.
- ☐ I can complete the 80% readiness checklist.
- ☐ I have written my error list before moving on.

50 fast recall prompts

1. Define market power.
2. Define product differentiation.
3. Give two examples of differentiated products.
4. Why does the demand curve slope downward?
5. What are the three conditions for demand?
6. What does WTP mean?
7. How do you build total demand from individual demand?
8. What changes quantity demanded?
9. What shifts demand?
10. Define total revenue.
11. Define total cost.
12. Define profit.
13. Write the constant-unit-cost profit formula.
14. Why can a firm not choose high P and high Q freely?
15. What are isoprofit curves?
16. Are isoprofit curves assessed in ECO114 Part A?
17. Define average cost.
18. Define marginal cost.
19. How do you identify fixed cost?
20. Why is AC undefined at $Q=0$?
21. Why can AC be U-shaped?
22. What happens to AC when $MC < AC$?
23. What happens to AC when $MC > AC$?
24. Derive MC from $TC = 1100 + 140Q + 0.2Q^2$.
25. Derive AC from $TC = 62500 + 500Q + Q^2$.
26. Find the MC at $Q = 1050$ for $TC = 62500 + 500Q + Q^2$.
27. At what Q is AC minimized for $TC = 62500 + 500Q + Q^2$?
28. Define marginal revenue.
29. Why is MR below P?
30. Explain the gain/loss method for MR.
31. If $P = 8000 - 80Q$, what is TR?
32. If $P = 8000 - 80Q$, what is MR?

33. What is the profit-maximising rule?
34. If $MR > MC$, what should the firm do?
35. If $MR < MC$, what should the firm do?
36. After finding Q^* , where do you read P^* ?
37. Define economies of scale.
38. Define diseconomies of scale.
39. Name three sources of economies of scale.
40. Explain technological economies.
41. Explain cost advantages.
42. Explain network economies.
43. Why might a firm become too large?
44. Define monopoly.
45. What is a natural monopoly?
46. Name three barriers to entry.
47. How does monopoly demand differ from differentiated-firm demand?
48. Write the profit rectangle formula.
49. Why is $P > MC$ for a price-setting firm?
50. What is the biggest Unit 7 Part A graph trap?

80% readiness checklist

☐ I can draw demand, MR and MC and find Q^* and P^* .

☐ I can explain why MR lies below demand.

☐ I can calculate TR, TC, AC, MC and profit from a table/function.

☐ I can distinguish movement along demand from a demand shift.

☐ I can explain product differentiation and market power.

☐ I can explain economies and diseconomies of scale with examples.

☐ I can distinguish monopolistic competition from monopoly.

☐ I can answer TF/MCQ without guessing recklessly.

☐ I can write a 6–10 mark answer on $MR=MC$.

☐ I know what is Part A only and what belongs to later Unit 7/8 packs.

17. One-page cram sheet

Section progress

0 / 3 (0%)

☐ I can use this sheet as a final pre-test checklist.

☐ I can recreate formulas and graphs from memory.

☐ I know the most dangerous traps.

Unit 7 Part A in one chain

Differentiation → market power → downward-sloping demand → MR below price → MR = MC gives Q^* → demand gives P^* → profit = $TR - TC$ or $Q(P - AC)$.

Must-know formulas

$TR = P \times Q$ | $\pi = TR - TC$ | $AC = TC/Q$ | $MC = \Delta TC / \Delta Q$ or dTC/dQ | $MR = \Delta TR / \Delta Q$ or dTR/dQ | If $P = a - bQ$, $MR = a - 2bQ$ | Profit max: $MR = MC$ | Profit rectangle:
 $\pi = Q(P - AC)$

Graph rules

- Demand slopes down.
- MR lies below demand for a price-setting firm.
- Q^* is where $MR = MC$.
- P^* is read from demand at Q^* .
- MC crosses AC at AC minimum.
- Economies of scale make LRAC fall; diseconomies make LRAC rise.

Traps

- Do not set $P = MC$ for a monopoly/price-setting firm.
- Do not read price from MR.
- Do not confuse AC with MC.
- Do not say market power means unlimited power.
- Do not call an own-price change a demand shift.
- Do not calculate profit with $Q(P - MC)$; use $Q(P - AC)$.